Practice workbook · Using a line's equation to find a missing coordinate or intercept

Working with a line whose equation (or gradient) is already known to find a missing coordinate, an axis crossing point, or to draw the line on a grid.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-COO-04

1. Question 9 · part a · 2 marks

2020 JAN · Paper 01 (regional R)

Worked-example retry · Attempt part a only.

9 The line l_1 with equation $y + 2x - 4 = 0$ passes through the point P with coordinates $(a, 6)$ and the point Q with coordinates $(3, b)$.

(a) Find the value of a and the value of b.

(2)

The line l_2 passes through point P and is perpendicular to l_1

The point R, with coordinates (e, f) lies on l_2 such that $PR = 6\sqrt{5}$

(b) Find the two possible pairs of values of e and f.

(8)

Given that $e < 0$,

(c) find the area of triangle PQR.

(3)

The points P, Q and R lie on a circle C.

(d) Find the coordinates of the centre of C.

(2)

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1. Question 9 · part a · 2 marks · Working and answer

Working and answer

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2. Question 10 · part a · 1 mark

2022 JAN · Paper 01

Further practice · Attempt part a only.

10 The equation of the line L_1 is $y - 2x - 6 = 0$

The point P with coordinates $(2, a)$ lies on L_1

(a) Find the value of a

(1)

The line L_2 is perpendicular to L_1 and passes through P

(b) Show that an equation of L_2 is $x + 2y - 22 = 0$

(4)

Line L_1 crosses the x-axis at the point A and line L_2 crosses the x-axis at the point B

The point C has coordinates (m, n) such that $m > 0$ and $n < 0$

The length of AC is $5\sqrt{2}$ and the gradient of BC is $\frac{1}{4}$

(c) Find the value of m and the value of n

(9)

(d) Find the area of quadrilateral $ACBP$

(3)

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2. Question 10 · part a · 1 mark · Working and answer

Working and answer

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3. Question 10 · part a · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part a only.

10 The equation of the line L_1 is $y - 3x + a = 0$

The point A with coordinates $(a, 10)$ lies on L_1

(a) Show that $a = 5$

(2)

Line L_1 crosses the y-axis at the point B

(b) Write down the coordinates of B

(1)

The point C with coordinates $(50, -5)$ lies on L_2

Given that L_2 passes through A

(c) (i) show that L_1 and L_2 are perpendicular

(3)

(ii) hence find an equation for L_2 giving your answer in the form $y = px + q$

(2)

The point D has coordinates (m, n)

The length of CD is $15\sqrt{10}$ and the gradient of BD is -3

(d) Find the value of m and the value of n

(6)

(e) Find the area of quadrilateral $ABCD$

(3)

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P 7 8 9 6 4 A 0 3 0 3 6

3. Question 10 · part a · 2 marks · Working and answer

Working and answer

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4. Question 10 · part b · 1 mark

2025 NOV · Paper 01

Further practice · Attempt part b only.

10 The equation of the line L_1 is $y - 3x + a = 0$

The point A with coordinates $(a, 10)$ lies on L_1

(a) Show that $a = 5$

(2)

Line L_1 crosses the y-axis at the point B

(b) Write down the coordinates of B

(1)

The point C with coordinates $(50, -5)$ lies on L_2

Given that L_2 passes through A

(c) (i) show that L_1 and L_2 are perpendicular

(3)

(ii) hence find an equation for L_2 giving your answer in the form $y = px + q$

(2)

The point D has coordinates (m, n)

The length of CD is $15\sqrt{10}$ and the gradient of BD is -3

(d) Find the value of m and the value of n

(6)

(e) Find the area of quadrilateral $ABCD$

(3)

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4. Question 10 · part b · 1 mark · Working and answer

Working and answer

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5. Question 8 · part c · 1 mark

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

8 The points A and B have coordinates $(1, 5)$ and $(9, 9)$ respectively.

(a) Find an equation of line AB, giving your answer in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(3)

The line l is perpendicular to AB and passes through the point X which lies on AB such that $AX : XB = 3:1$

(b) Show that an equation of l is $y = -2x + 22$

(5)

The point C has coordinates $(6, p)$

Given that C lies on l

(c) find the value of p

(1)

$ABCD$ is a parallelogram where the x coordinate of D is negative.

(d) Find the coordinates of the point D

(3)

(e) Find the area of the parallelogram $ABCD$

(4)

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P 7 4 2 8 4 A 0 2 0 3 2

5. Question 8 · part c · 1 mark · Working and answer

Working and answer

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6. Question 4 · part a · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part a only.

4 The point A with coordinates $(12, 14)$ and the point B with coordinates $(q, 2)$ where q is a constant, lie on the straight line with equation $3y - 2x - p = 0$ where p is a constant.

(a) Find the value of p and the value of q

(3)

The line L is perpendicular to AB and passes through the point X, which lies on AB such that $AX : XB = 1 : 2$

(b) Find an equation for L in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(6)

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6. Question 4 · part a · 3 marks · Working and answer

Working and answer

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7. Question 5 · part a · 2 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

5 The line l with gradient $-\frac{1}{12}$ passes through the points A and B with coordinates $(p, 10)$ and $(123, 0)$ respectively.

(a) Show that $p = 3$

(2)

(b) Find an equation for l in the form $rx + sy + t = 0$ where r, s and t are integers.

(2)

The line k is perpendicular to l and passes through the point A.

(c) Find an equation for k in the form $y = mx + c$

(3)

Line k intersects the x-axis at the point C.

(d) Find the exact area of triangle ABC.

(4)

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7. Question 5 · part a · 2 marks · Working and answer

Working and answer

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